



Fig. 1: Structure of light-emitting diode



Fig. 2: Correct way to connect an LED

LEDs can be damaged by heat while soldering. So you have to be quick enough to ward off this small risk. Never connect an LED directly to a battery or power supply. It will

destroy almost instantly because too much current will pass through and burn it out.

LEDs must have a resistor in series to limit the current to a safe value. For the purpose of quick testing, a 1-kilo-ohm resistor is suitable for most LEDs if your supply voltage is 12V or less.

Materials used for production

LEDs are usually built on an n-type substrate, with an electrode attached to the p-type layer deposited on its surface. Many commercial LEDs, especially GaN/InGaN, also use sapphire substrate.

Most materials used for LED production have very high refractive indices. This means most of the light reflects back into the material at the material/air interface. Therefore light extraction is an important aspect of LED design and production. It calls for a lot of R&D work.

Conventionally, LEDs are made from a variety of inorganic semiconduc-

TABLE-I
Semiconductor Materials Used for Different Colour LEDs

Colour	Wavelength (nm)	Voltage (V)	Semiconductor material
Infrared	$\lambda > 760$	$\Delta V < 1.9$	Gallium arsenide (GaAs) Aluminium gallium arsenide (AlGaAs)
Red	$610 < \lambda < 760$	$1.63 < \Delta V < 2.03$	Aluminium gallium arsenide (AlGaAs) Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Orange	$590 < \lambda < 610$	$2.03 < \Delta V < 2.10$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Yellow	$570 < \lambda < 590$	$2.10 < \Delta V < 2.18$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Green	$500 < \lambda < 570$	$2.18 < \Delta V < 4.0$	Indium gallium nitride (InGaN) / Gallium(III) nitride (GaN) Gallium(III) phosphide (GaP) Aluminium gallium indium phosphide (AlGaInP) Aluminium gallium phosphide (AlGaP)
Blue	$450 < \lambda < 500$	$2.48 < \Delta V < 3.7$	Zinc selenide (ZnSe) Indium gallium nitride (InGaN) Silicon carbide (SiC) as substrate Silicon (Si) as substrate—under development
Violet	$400 < \lambda < 450$	$2.76 < \Delta V < 4.0$	Indium gallium nitride (InGaN)
Purple	Multiple types	$2.48 < \Delta V < 3.7$	Dual blue/red LEDs, blue with red phosphor, or white with purple plastic
Ultraviolet	$\lambda < 400$	$3.1 < \Delta V < 4.4$	Diamond (C) Aluminium nitride (AlN) Aluminium gallium nitride (AlGaIn) Aluminium gallium indium nitride (AlGaInN)—down to 210 nm
White	Broad spectrum	$\Delta V = 3.5$	Blue/UV diode with yellow phosphor

tor materials. Table I shows the available colours with wavelength range, voltage drop and material used.

Why use LEDs for lighting?

After going through the following parameters of LED lights, you'll find that LEDs are the best solution for energy-efficient and pollution-free lighting.

Lifetime. As a solidstate light source, LEDs have a very long lifetime and are generally very robust due to no mechanical or moving parts. Incandescent bulbs have an expected lifetime of 1000 to 5000 hours, while good-quality LEDs can last up to 50,000 hours—more than five years of continuous use.

Low maintenance. The long lifetime of LEDs reduces the need to replace failed lamps, and this can lead to

significant financial and environmental savings, particularly in maintenance, labour and recycling.

Efficiency. As a semiconductor device, LEDs are highly efficient. Current device technology is allowing their performance to be pushed to limits that exceed that of standard lighting. The directional nature of light produced by LEDs allows the design of luminaries with higher overall efficiency.

Low power consumption. The low power consumption of LEDs leads to large energy savings.

Brightness. Light outputs of LEDs are constantly being increased, and there are several products that have higher lumen output than traditional lighting.

Heat. Almost 80 per cent of the energy consumed by LEDs is converted